

# Apartment Prices Prediction Project

For this assignment, you will predict the price of several apartments using basic Machine Learning techniques. You can use any algorithm of the first ones we will discuss in class. We especially encourage you to try algorithms like Lasso and Ridge regression, and compare their coefficients to those of a standard linear regression (and amongst themselves). You will use the data and Kaggle page from:

<https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques>

You will need to create a Kaggle account or use an existing one. Then, you will need to join the competition and set up your team's name.

It is of **utmost importance** that you put your NIS (the number that appears on the backside of your BSE student ID, but without any letters, only the numbers). This is because the leaderboard already has many teams, so we need a fast way to find your individual scores. **Whoever fails to correctly set up their team's name will not receive any grade** for the Score obtained for their submissions.

Download the competition data from Kaggle, then build a Jupyter Notebook with your pipeline (a simple model is fine). The notebook should output a .csv file containing your SalePrice predictions for test.csv, following the competition's required format (you can check the provided sample "test.csv" file). Finally, submit that file on Kaggle using the "Submit Predictions" button. This will yield you a score on the competition metric (RMSE). Beware that you can make at most 10 submissions per day, so don't use them up testing minor tweaks. Save them for changes that are likely to make a real difference.

## Your submission for this assignment is:

1. A Jupyter Notebook with all your code and the block that creates your predictions as a .csv file.
2. Score on the leaderboard of the aforementioned Kaggle competition.

If you have trouble downloading the data or have any other questions about the project, Guillem will solve your doubts at the first TA on October 8<sup>th</sup>.

The project must be submitted INDIVIDUALLY, although we encourage you to work together on it to contrast methodologies and discuss your results with your classmates to improve your learning experience and gain better ideas.

**DEADLINE** for the Classroom submission of your notebooks: October 28<sup>th</sup> @ 17:00

### EVALUATION:

We will give you a grade based on two main areas:

- Score on Kaggle
  - Don't stress about this; the score is not ranked in the class. We will only evaluate whether you scored better than a simple benchmark. The benchmark is easy to beat. Any reasonable model built with the methods and principles from class should score higher.
  - We may or may not disclose the benchmark score before the deadline, depending on how the competition evolves.
- Modelling choices in the Jupyter Notebook
  - We will investigate your modelling choices and check some boxes. We have some things in mind that you should be able to do. If you do them, you get full marks for this.
- (OPTIONAL) Ex-post discussion during the TA session on October 30<sup>th</sup>
  - Guillem will hold a TA session where he'll give feedback on your notebooks (anonymized if anything is too specific). You can also ask him how to improve for future assignments, or any other questions about the project.
  - Active participation that showcases a minimum level of understanding of your modelling choices will be rewarded with up to an additional point in the final grade.
  - The threshold to getting the point will also be very low. Just go and discuss your modelling choices if you want to get it. You won't be judged for your model.

### GRADING:

- Jupyter Notebook + Score on Kaggle: grade out of 10
  - Weight in the final grade: 15%
- Participation in the **optional** discussion with the TA: point or no point.
  - Weight in the final grade: NA
  - You can get a 10/10 in the final grade without those additional points, but they are a whole point. If you get a 7/10 as your final grade after the

assignments and the final project have been graded, then +1 point from the TA means you get an 8/10 as your final grade.